

*MA.6.AR.3.5***Benchmark**

MA.6.AR.3.5 Solve mathematical and real-world problems involving ratios, rates and unit rates, including comparisons, mixtures, ratios of lengths and conversions within the same measurement system.

Benchmark Clarifications:

Clarification 1: Instruction includes the use of tables, tape diagrams and number lines.

Connecting Benchmarks/Horizontal Alignment

- MA.6.NSO.1.1
- MA.6.NSO.2.1, MA.6.NSO.2.3
- MA.6.NSO.3.1, MA.6.NSO.3.5
- MA.6.AR.1.1
- MA.6.AR.2.3
- MA.7.AR.3.1, MA.7.AR.3.2

Terms from the K-12 Glossary

- Customary units
- Metric units
- Rate
- Unit Rate

Vertical Alignment**Previous Benchmarks**

- MA.5.M.1.1
- MA.5.FR.1.1
- MA.5.FR.2.1, MA.5.FR.2.2
- MA.5.AR.1.2, MA.5.AR.1.3

Next Benchmarks

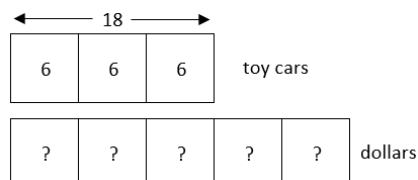
- MA.7.AR.3.3

Purpose and Instructional Strategies

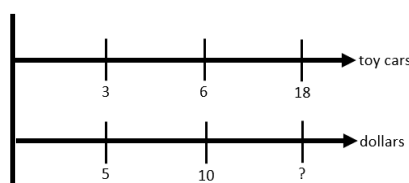
In previous courses, students developed an understanding of fraction concepts and operations, often in the context of the relationship that relates parts to wholes. Students also learned how to convert measurements within the same measurement system. In grade 6 accelerated, students solve problems involving ratio relationships. Students will also solve problems involving proportional relationships. This benchmark is the culmination of the MA.6.AR.3 standard. It combines all of the concepts developed in the previous four benchmarks and applies them to a variety of mathematical and real-world contexts, including conversions of units within a measurement system. In future courses, students are expected to convert units across different measurement systems.

- Instruction includes a variety of problem types including comparison, mixtures, lengths and conversions. Students should work within the same measurement system, for example, standard measurement to standard measurement or metric system to metric system.
- It is not the expectation of this benchmark to solve problems using proportions; however, problem types will naturally start to build the foundation of proportions and proportional relationships.

- Instruction includes the use of bar models, tables and number lines to help students see relationships and relevance to real-world applications (*MTR.2.1*).
 - For example, 3 toy cars cost Mikela \$5 at the local superstore. Mikela needs 18 toy cars. How much will it cost Mikela to buy the cars?
 - Bar Models



- Number Lines



- Tables

Toy cars	3	6	18
dollars	5	10	?

- Discussion about why students used certain methods to solve specific problems can help students see the patterns for determining their own processes in the future (*MTR.2.1*, *MTR.5.1*, *MTR.7.1*).
- Instruction includes many opportunities for converting between units in the same measurement system.

Common Misconceptions or Errors

- Some students may struggle to determine a process for solving problems in a real-world context. Instruction highlights strategies for students to build perseverance by modifying methods as needed and analyzing the problem in a way that makes sense (*MTR.1.1*).

Strategies to Support Tiered Instruction

- Teacher provides opportunities for students to comprehend the context or situation by engaging in questions (writing questions on flashcards to reuse will be useful) such as:
 - What do you know from the problem?
 - What is the problem asking you to find?
 - Can you create a visual model to help you understand or see patterns in your problem?
- Instruction includes using colored pencils to identify the units in each corresponding portion of a ratio and identifying the units before writing the numerical values.

- Instruction includes the use of a three-read strategy. Students read the problem three different times, each with a different purpose.
 - First, read the problem with the purpose of answering the question: What is the problem, context, or story about?
 - Second, read the problem with the purpose of answering the question: What are we trying to find out?
 - Third, read the problem with the purpose of answering the question: What information is important in the problem?
- Instruction highlights strategies for students to build perseverance by modifying methods as needed and analyzing the problem in a way that makes sense.

Instructional Tasks

Instructional Task 1 (MTR.7.1)

On the first day of camp, Sara hiked 20 miles in 5 hours.

Part A. At this rate, how long would it take Sara to hike 12 miles? Show and explain how you determined this.

Part B. At this same rate, how many miles could Sara hike in 2 hours? Show and explain how you determined this.

Instructional Task 2 (MTR.7.1)

Chris and Jenny are comparing two similar punch recipes. Each recipe calls for cranberry juice and ginger ale, but in different amounts. The tables below show the amounts of cranberry juice and ginger ale for four different quantities of punch.

Chris's Punch	
Cranberry Juice (in cups)	Ginger Ale (in cups)
1	4
2	8
3	12
5	20

Jenny's Punch	
Cranberry Juice (in cups)	Ginger Ale (in cups)
2	3
4	6
6	9
10	15

Part A. Is the ratio of the punch that is cranberry juice the same in each of Chris's recipes given in his table? Explain how you determined your answer.

Part B. Is the ratio of the punch that is cranberry juice the same in each of Jenny's recipes given in her table? Explain how you determined your answer.

Part C. Is the ratio of the punch that is cranberry juice the same in Chris's recipes as it is in Jenny's recipes? If not, whose punch has a greater concentration of cranberry juice? Explain how you determined your answer.

Instructional Items

Instructional Item 1

A recent study found that parking lots for offices should have a ratio of 6 spaces for every 1000 square feet of floor space. If a new office building has 19,000 square feet of floor space, how many spaces should there be in the parking lot?

Instructional Item 2

Jessica made 8 out of 24 free throws. Bob made 5 out of 20 free throws. Who has the highest free-throw ratio?